

Mathematical Statistics – Final Project Guidelines

Deadline: To be announced after the midterm exam.

Overview

The final project is an **individual assignment** designed to encourage independent research, critical thinking, and the practical application of statistical theory. **Each student** is required to submit a comprehensive report of **10–15 pages** in PDF format.

As this is an individual endeavor, students are expected to produce original work that reflects their own understanding and analysis of the chosen topic.

Students may **choose one** of the following three directions for their project:

1. Data Analysis

Conduct an empirical analysis on a dataset of your choice. Your report should include:

- A detailed description of the data source and variables.
- Clearly defined target **research question(s)**.
- The statistical **methodology** adopted (e.g., estimation, hypothesis testing, or model selection).
- **A discussion of results**, diagnostic checks, and conclusions.
- *Note:* Pay close attention to the structure and clarity of your writing to ensure the analysis is reproducible and understandable.

2. Methodological Innovation & Extension

Present an original idea or innovation based on modern statistical literature.

- **Source Requirement:** You must select a statistical method published within the **past five years (2021–2026)** in one of the following journals: *JASA*, *Annals of Statistics*, *Biometrika*, *JRSSB*, *JMLR*, *Econometrica*, *Review of Economic Studies*, *Journal of Econometrics*, *JBES*, or *Econometric Theory*. **Other high-quality journals may be allowed with the prior permission of the instructor.**
- **The Task:** Propose a **modification, extension, or a new perspective on the chosen method**. This could involve relaxing specific assumptions, improving computational efficiency, or adapting the method for a different data structure.
- **Validation:** **Demonstrate the performance of your proposed innovation via simulation studies**. While formal theoretical proofs are optional, you must clearly articulate **the intuitive logic and mathematical framework** of your method. Supporting your work with real-world data is encouraged but not required.

3. Statistical Foundations of Large Language Models (LLMs)

Introduce and analyze a learning method or algorithm used in the context of Large Language Models.

- **The Task:** Select a **specific algorithm or technique** (e.g., Attention mechanisms, RLHF, Low-Rank Adaptation (LoRA), or Quantization methods).
- **Statistical Analysis:** Explain the **core statistical concepts or ideas underlying the method** (e.g., high-dimensional probability, optimization theory, or Bayesian inference).

- **Demonstration:** You must demonstrate the effectiveness of the method through **at least one** of the following:
 1. **Simulation Examples:** Replicating the algorithm's behavior in a controlled setting.
 2. **Empirical Applications:** Applying the method to a specific task to show performance gains or qualitative shifts.
- **Critical Insights:** Include your own analysis regarding the statistical limitations, robustness, or future potential of the chosen approach.

Submission Requirements

Each student must upload a compressed package (.zip or .rar) to the digital teaching platform. The package should include:

1. **Project Report:** A single PDF file (10–15 pages).
2. **Code File:** All scripts used for simulations or data analysis (e.g., .R, .py, or .m).
3. **Data File:** Required only if your project involves real-world data analysis (not needed for simulation-only projects).

File Naming Convention:

Please name your compressed file as: YourName+StudentID (e.g., JohnDoe+20260011.zip).